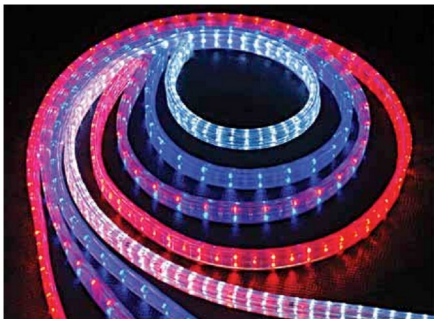


Ensuring LEDs' Long Life

LEDs have an extremely long operating life of 50,000 hours—equating to seventeen years at eight hours daily. By then the lamp is amortised. But what happens when the driver packs up after only a few years' operation?



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Long before Fukushima, energy saving was a watchword. Efficient use of energy reduces the number of power stations. The conventional filament lamps are problematic because of their low efficiency. On the other hand, energy-saving CFL lamps too are not without problems due to their mercury content. For this reason, advances in light-intensive white LEDs are welcome. These offer an additional advantage—long life. With proper cooling, LED lamps have a life expectancy 50 times greater than filament lamps. However, these might not be within design trends when they finally reach the end of their life-cycle!

With life expectancy approaching 50,000 hours for LED lighting, their service cost tends toward zero. One only has to consider the cost involved in replacing

the lamp. The introduction of electronics into lighting technology has transformed it into a system, which offers energy savings along with substantial reductions in service cost. This is conditional on a balanced adjustment of the LED, the driver and the cooling. This system, however, is only as good as its weakest link.

Electronics posing challenges for designers

Designing highly efficient LED-based lighting systems confronts the designers with electronics issues. LEDs are semiconductors that operate with low DC voltages. To turn these into lights with a high life expectancy, designers have to pay attention to selecting the optimal LEDs and combining these with the right power supply and optimised cooling. The individual components determine not only the quality of the emitted light (what we see and feel) but

also the efficiency and life expectancy of the complete system.

Cooling—the central aspect

It is widely believed that LEDs do not heat up. In fact, these radiate very little heat, quite in contrast to filament lamps. Converting electrical current into light generates heat in the confined barrier (depletion) layer. So cooling or heat management is an important aspect as high temperatures in the depletion barrier have a negative impact on the life of the lamp. Heat sinks must be sufficiently big, so that the temperature on the surface of the LEDs doesn't exceed +75°C.

Good heat management and maintaining the temperature of the depletion layer continuously at 80°C makes it possible to increase the life expectancy towards the magic 100,000 hours limit (Fig. 1). An additional advantage is that a maximum temperature of +55°C on the heat sink of a lamp poses no injury threat to the user. Meanwhile there are a multitude of heat sinks in various forms and sizes available, which leave the designers with many options.

Fig. 2 shows the current-voltage curve of high-power LEDs. If you choose a supply voltage of, say, 3V LED1 is over-driven, LED2 draws 300 mA, LED3 draws 250 mA and LED4 draws only 125 mA. Furthermore, these curves are dynamic. As the LEDs warm up to their operating temperatures, all curves drift to the left (the forward voltage (V_F) reduces with increasing temperature).

The driver is critical

Although LEDs have been around for decades, their use for lighting has been discovered only recently. Quite contrary to display applications with in-